

Federico Salerno

Postdoctoral Researcher · Politecnico di Torino · 3D Lab, DIGEP

federico.salerno@polito.it · orcid.org/0000-0001-5282-9513 · linkedin.com/in/f-salerno · Torino, Italy

RESEARCH INTERESTS

Augmented Surgery · Computer-Assisted Surgery · Mixed Reality (AR/MR) · Surgical Navigation · 3D Reconstruction · Medical Imaging · Optical Tracking · Biomedical System Integration

EXPERIENCE

Postdoctoral Researcher

Nov 2025 – Present

Politecnico di Torino · 3D Lab, DIGEP

Markerless optical tracking for augmented surgical applications, gesture-based interaction modeling, real-time pose estimation, spatial registration robustness, and experimental validation protocols for AR-guided procedures.

PhD Student

Nov 2022 – Nov 2025

Politecnico di Torino · 3D Lab, DIGEP

PhD research in augmented reality for surgery. Design of AR solutions to register virtual holographic anatomical models onto the operating field. Development of methodologies for accuracy and precision evaluation of AR tracking systems and optical tracking benchmarking.

Visiting PhD Student

Oct 2024 – Mar 2025

Technische Universität Wien · Vienna, Austria

Research visit focused on augmented reality applications for surgical guidance.

Research Fellowship

May 2022 – Nov 2022

Politecnico di Torino

Dimensional and mechanical validation of surgical cutting guides. Geometric distortion analysis via 3D scanning. FEM analysis and experimental tensile testing.

CAD-CAM Prototyping

May 2021 – Oct 2021

Dropper S.r.l. · Torino

CAD design of a people counter device casing.

CAD-CAM Prototyping

Nov 2019 – Sep 2021

Nova Stark S.r.l. · Torino

CAD design of agritech device casing for viticulture environmental parameter measurement. Device prototyping and experimental test bench for sensor testing.

Video Lectures on Additive Manufacturing

Apr 2021 – May 2021

Sin Tesi Forma S.r.l. · Torino

Pre-recorded training course (3 h) on additive manufacturing in the industry.

Research Fellow (Part-time)

Oct 2019 – Jan 2020

Politecnico di Torino

Assistance during Industrial Technical Drawing lab sessions.

EDUCATION

Doctor of Philosophy (PhD) — Cum Laude

2022 – 2025

Politecnico di Torino

Thesis: "Augmented Surgery Solutions"

Master's Degree, Mechanical Engineering (Mechanical Production) — 110L/110

2019 – 2022

Politecnico di Torino

Bachelor's Degree, Mechanical Engineering — 107/110

2016 – 2019

SELECTED PUBLICATIONS

Advancing surgical cutting guide flexibility: A hybrid physical and Augmented Reality solution for cranio-maxillofacial surgery

Salerno F., Ulrich L., Marullo G., Moos S., Vezzetti E. · *Journal of Cranio-Maxillofacial Surgery* (2026) · DOI: 10.1016/j.jcms.2026.104480

A metrological approach for Augmented Reality tooltip tracking assessment

Salerno F., Ulrich L., Maculotti G., Moos S., Genta G., Vezzetti E., Galetto M. · *Computers in Industry* (2026) · DOI: 10.1016/j.compind.2025.104430

A 3D Camera-Based Approach for Real-Time Hand Configuration Recognition in Italian Sign Language

Ulrich L., De Luca A., Miraglia R., Mulassano E., Quattrocchio S., Marullo G., Innocente C., Salerno F., Vezzetti E. · *Sensors* (2026) · DOI: 10.3390/s26031059

MOTT: Modular Optical Tool Tracking Framework enabling Efficient Benchmarking

Salerno F., Contenti A., Ulrich L., Marullo G., Moos S., Vezzetti E. · *Scientific Reports* (2025) · DOI: 10.1038/s41598-025-21044-z

A Methodology for the Dimensional and Mechanical Analysis of Surgical Guides

Salerno F., Moos S., Ulrich L., Novaresio A., Vezzetti E. · *Lecture Notes in Mechanical Engineering* (2024) · DOI: 10.1007/978-3-031-52075-4_22

How to exploit Augmented Reality (AR) technology in patient customized surgical tools: a focus on osteotomies

Ulrich L., Salerno F., Moos S., Vezzetti E. · *Multimedia Tools and Applications* (2024) · DOI: 10.1007/s11042-023-18058-y

Development of Bleeding Artificial Intelligence Detector (BLAIR) System for Robotic Radical Prostatectomy

Checucci E., Piazzolla P., Marullo G., Innocente C., Salerno F., Ulrich L., Moos S., et al. · *Journal of Clinical Medicine* (2023) · DOI: 10.3390/jcm12237355

LICENSES & CERTIFICATIONS

Abilitazione all'esercizio della professione di ingegnere industriale (Dec 2022)

Ordine degli Ingegneri di Torino